Systems Thinking Challenge

Understand → Model → Analyze → Decide → Justify

- **2** Problem identification & OS concept recognition
- **2** Problem modelling
- **2** Analysis of alternatives
- **2** Selection of appropriate mechanism
- **2** Engineering justification & communication

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MARKS

THINK

Critical Thinking + Research Experience

Performance Investigation

Hypothesis → Experiment → Measure → Compare → Conclude

- **2** Experimental design & methodology
- **2** Implementation & correctness
- **2** Data collection & measurement
- **2** Analysis & visualization
- **2** Engineering interpretation & conclusion

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MARKS

MEASURE

Critical Thinking + Innovation & Prof. Development

Design & Debugging Challenge

Diagnose → Design → Implement → Validate → Defend

2	Problem diagnosis & root cause
2	System / algorithm modelling & reasoning
2	Solution design & alternative analysis
2	Implementation & validation
2	Technical defence & trade-offs

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MARKS

ENGINEER

Critical Thinking + Innovation & Prof. Development

Research Mini-Project

Question → Literature → Experiment → Analyze → Conclude

2	Research problem, question & gap
1	Literature review & justification
2	Experimental methodology & design
1	Prototype / implementation
2	Data collection, analysis & visualization
1	Interpretation, limitations & conclusion
1	Communication & AI-use disclosure

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MARKS

INVESTIGATE

Research Experience + Critical Thinking

Linux Systems Engineering

Observe → Instrument → Intervene → Validate → Defend

2	Problem & OS mechanism understanding
2	Linux instrumentation & tool usage
2	Implementation / engineering intervention
2	Experimental evidence & analysis
2	Validation, explanation & defence

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MARKS

OBSERVE

Innovation & Prof. Development + Real World Exposure

Rust Micro-OS Project

Boot → Tasks → Scheduler → Sync → Memory → Integrate

2	Architecture & OS mechanism design
4	Rust implementation quality
3	Core OS functionality
2	Testing, debugging & validation
2	Performance / evidence-based evaluation
1	Innovation / specialization
1	Documentation & demonstration

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MARKS

BUILD

Innovation & Prof. Development + Critical Thinking

Viva & Demonstration

Demonstrate → Explain → Modify → Diagnose → Justify

1. **1** Technical demonstration
2. **1** Conceptual understanding
3. **1** Engineering reasoning & trade-offs
4. **1** Debugging / scenario-based questioning
5. **1** Communication, contribution & defence

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MARKS

DEFEND

Critical Thinking